

Carl Louis

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EDUCATION

Singapore Institute of Technology | University of Glasglow

Sep 2023 – May 2026

Bachelor of Engineering with Honours in Aerospace Engineering – Cumulative GPA: 4.14/5.00

Cambridge International AS & A Level

Jun 2022

- Mathematics (A), Physics (A), Chemistry (A), Biology (B)

EXPERIENCE

Addvalue Technologies

May 2025 – Dec 2025

Project Engineer Intern

- Supported BOM traceability, engineering change documentation, and cross-team coordination between mechanical, RF, and reliability engineering team.
- Developed thermal modelling and analysis for space communication module, evaluating conduction and radiation paths to support thermal specification design decisions.
- Assisted in reliability testing such as vibration and radiation susceptibility.

Kairos Gracia Christian School

Jul 2022 – Dec 2022

Lab Assistant Intern

- Managed labs and designed activities practise for O-level and A-level practical tests preparation.

PROJECTS

Personal Wing-In-Ground Craft Development

- Designed a conceptual prototype targeting low-altitude flight over water using aerodynamic ground effect principles.
- Performed CFD analysis in ANSYS Fluent to evaluate airfoil performance for ground-effect flight regimes.
- Fabricated foam core prototype airframe.

SIT Airborne

Avionics, Navigation and Controls Student Engineer

- Led the design and integration of flight controllers and avionics components of the large drone prototype.

AWARDS / ACHIEVEMENTS

ASEAN Undergraduate Scholarship

Apr 2023

TECHNICAL SKILLS

- Programming: Python, MATLAB
- Engineering: ANSYS Mechanical, OpenFOAM, SolidWorks, Fusion 360
- Data Analysis: NumPy, Pandas, SciPy, Excel

LEADERSHIP AND INVOLVEMENT EXPERIENCE

International Students Network (SIT)

Jan 2024 – Dec 2024

Secretary

- Drafted necessary documents for activities/events and coordinated meeting agenda.
- Budgeted funds for activities/events.

Kickboxing (SIT)

Jan 2024 – Dec 2024

Head of Logistics

- Managed logistics such as inventory and venues for trainings and events.